

MEHRABAD INTL, Iran, Islamic Republic ofWMO: **407540**Lat: **35.689N**Lon: **51.313E**Elev: **3962**StdP: **12.71**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	29.8	7.0	9.1	48.0	10.5	10.9	47.8	27.1	46.3	23.5	46.7	5.6	280	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	102.2	63.7	100.2	63.4	98.2	63.2	71.1	90.8	68.4	91.9	66.7	90.9	8.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	104.7	86.5	59.6	88.3	82.7	57.1	80.7	82.5	37.7	90.2	35.2	92.0	33.6	91.3	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	21.0	17.1	22.4	105.1	4.6	2.0	19.1	106.5	16.4	107.7	13.8	108.8	10.4	110.2		
(5)			19.9	69.6	3.3	3.9	17.5	72.4	15.6	74.7	13.7	76.9	11.3	79.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	41.3	45.1	53.8	63.6	73.8	83.4	87.9	86.7	79.2	67.8	53.2	44.4	(6)	(7)
	DBStd	17.58	6.42	6.88	7.75	7.07	5.83	5.06	4.40	3.86	5.00	6.28	7.01	5.57	(7)	(8)
	HDD50	720	275	158	48	5	0	0	0	0	0	0	45	189	(8)	(9)
	HDD65	2797	735	557	355	105	6	0	0	0	0	45	355	639	(9)	(10)
	CDD50	6247	6	22	167	412	737	1002	1176	1139	877	552	142	15	(10)	(11)
	CDD65	2848	0	0	9	61	279	553	711	674	427	132	2	0	(11)	(12)
	CDH74	37845	0	0	45	402	2668	7800	11061	10070	4901	895	3	0	(12)	(13)
	CDH80	20771	0	0	4	78	929	4371	6989	6005	2198	197	0	0	(13)	(14)
(14) Wind	WSAvg	6.4	5.2	6.6	7.8	7.8	8.5	7.5	6.9	6.0	5.9	5.6	4.8	4.4	(14)	(15)
(15) Precipitation	PrecAvg	8.5	1.3	1.1	1.4	1.3	0.6	0.1	0.1	0.1	0.1	0.4	0.9	1.3	(15)	(16)
	PrecMax	15.3	4.6	4.3	5.6	5.2	2.0	0.5	0.6	0.9	0.9	1.9	3.1	3.1	(16)	(17)
	PrecMin	3.9	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)	(18)
	PrecStd	3.3	1.0	0.8	1.3	1.1	0.5	0.2	0.1	0.2	0.2	0.5	0.7	0.8	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	64.8	79.1	86.3	93.3	102.0	105.5	103.5	96.7	87.8	73.5	62.9	(19)	(20)
		MCWB	44.8	45.8	51.4	56.3	59.1	63.0	64.4	64.3	62.2	57.4	50.7	47.5	(20)	(21)
	2%	DB	56.8	60.8	73.5	82.2	90.0	98.9	103.3	100.6	94.0	84.5	69.9	58.9	(21)	(22)
		MCWB	43.0	44.4	50.7	56.3	58.4	62.4	64.0	63.8	60.9	56.1	50.4	45.0	(22)	(23)
	5%	DB	53.6	58.3	69.7	78.6	87.8	96.9	100.8	98.7	91.8	82.2	66.6	55.5	(23)	(24)
		MCWB	41.6	43.5	49.1	55.3	58.0	61.5	63.4	63.8	60.4	55.6	49.2	43.9	(24)	(25)
	10%	DB	50.7	55.5	66.1	75.3	85.6	94.9	98.9	96.8	89.8	79.0	64.1	53.2	(25)	(26)
		MCWB	40.4	42.2	47.3	53.8	57.2	60.7	63.4	63.2	59.9	54.8	48.3	42.6	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	49.1	58.3	64.4	67.0	70.4	73.9	76.0	72.4	62.7	55.6	49.2	(27)	(28)
		MCDB	54.3	59.2	68.5	76.3	85.4	94.0	94.4	92.2	88.6	77.9	65.2	59.5	(28)	(29)
	2%	WB	45.3	46.7	53.2	59.8	62.3	67.0	70.4	71.8	68.1	60.0	53.0	46.8	(29)	(30)
		MCDB	52.7	56.4	66.3	74.1	83.0	89.8	93.5	88.5	86.1	76.5	62.9	55.4	(30)	(31)
	5%	WB	43.3	45.2	51.0	57.2	60.3	64.8	68.4	67.5	64.7	58.3	51.2	45.5	(31)	(32)
		MCDB	50.7	55.4	65.4	72.8	81.8	89.1	93.3	92.8	85.2	75.8	62.5	52.9	(32)	(33)
	10%	WB	41.5	43.6	49.1	55.4	58.9	63.1	66.8	66.0	62.1	56.6	49.5	44.0	(33)	(34)
		MCDB	49.3	53.6	62.8	71.9	80.3	88.8	92.1	91.6	84.1	74.3	61.3	50.5	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	14.2	15.8	16.7	18.2	20.0	19.9	18.7	18.3	16.5	13.7	12.2	(35)	(36)
		MCDBR	15.0	16.1	18.0	18.4	19.6	20.5	20.7	19.6	19.1	18.6	16.0	15.4	(36)	(37)
	5% WB	MCWBR	7.8	8.1	7.9	7.3	6.8	6.2	5.7	6.0	7.0	6.8	6.6	7.9	(37)	(38)
		MCWBR	12.8	14.1	15.7	16.9	18.5	20.9	19.9	19.0	18.7	17.0	13.7	12.7	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.314	0.348	0.406	0.474	0.476	0.467	0.504	0.459	0.404	0.400	0.355	0.318	(39)	(40)
		taud	2.337	2.229	2.067	1.935	1.958	2.003	1.931	2.053	2.190	2.169	2.281	2.369	(40)	(41)
	Ebn at Noon		278	280	272	259	259	261	250	259	269	257	259	269	(41)	(42)
		Edh at Noon	30	38	49	59	59	56	60	52	43	40	32	28	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		840	1084	1435	1751	2141	2463	2358	2192	1834	1340	912	749	(43)	(44)
	RadStd		61	94	124	163	109	100	79	84	62	72	89	82	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-4.03	N/A	-2.84	-3.24	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.43	N/A	-1.40	+0.95	-0.65	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page